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1. Experiment Title:

Design a Registration Form for Gym Admission and use the appropriate Form Validation and Events of Javascript

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 8.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Registration Form for Gym Admission - Case Study 8</title>
    <link rel="stylesheet" href="Case Study 8.css">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
    <div class="container">
        <h2>Gym Admission Form</h2>

        <form id="gymForm">
            <table class="table-form">
                <thead>
                    <tr>
                        <th style="width: 38%;">Name</th>
                        <th style="width: 62%;">Value</th>
                    </tr>
                </thead>
                <tbody>
                    <tr>
                        <td class="label-cell">Name</td>
                        <td class="value-cell">
                            <input type="text" id="name" name="name" placeholder="Enter your name">
                            <div class="error-msg" id="nameError"></div>
                        </td>
                    </tr>
                    <tr>
                        <td class="label-cell">Sex</td>
                        <td class="value-cell">
                            <div class="radio-group">
                                <label class="radio-item">
                                    <input type="radio" name="sex" value="Male"> Male
                                </label>
                                <label class="radio-item">
                                    <input type="radio" name="sex" value="Female"> Female
                                </label>
                            </div>
                            <div class="error-msg" id="sexError"></div>
```

```
                        </td>
                    </tr>
                    <tr>
                        <td class="label-cell">Eye color</td>
                        <td class="value-cell">
                            <select id="eyeColor" name="eyeColor">
                                <option value="green">green</option>
                                <option value="blue">blue</option>
                                <option value="brown">brown</option>
                                <option value="black">black</option>
                                <option value="hazel">hazel</option>
                            </select>
                            <div class="error-msg" id="eyeColorError"></div>
                        </td>
                    </tr>
                    <tr>
                        <td class="label-cell">Check all that apply</td>
                        <td class="value-cell">
                            <div class="checkbox-group">
                                <label class="checkbox-item">
                                    <input type="checkbox" id="over6Feet" value="Over 6 feet tall"> Over 6
feet tall
                                </label>
                                <label class="checkbox-item">
                                    <input type="checkbox" id="over200Pounds" value="Over 200 pounds">
Over 200 pounds
                                </label>
                            </div>
                        </td>
                    </tr>
                    <tr>
                        <td colspan="2" class="full-row-header">Describe your athletic ability:</td>
                    </tr>
                    <tr>
                        <td colspan="2" class="textarea-cell">
                            <textarea id="ability" rows="5" placeholder="Enter details about your athletic
ability..."></textarea>
                            <div class="error-msg" id="abilityError"></div>
                        </td>
                    </tr>
                    <tr>
                        <td colspan="2" class="action-row">
                            <button type="submit" id="submitBtn">Enter my information</button>
                        </td>
                    </tr>
                </tbody>
            </table>

            <div id="resultCard" class="summary-card"></div>
        </form>

        <footer>
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Case Study 8.js"></script>
</body>
</html>
```

Case Study 8.css

```
* {
    box-sizing: border-box;
}

body {
    background-color: whitesmoke;
    color: black;
```

```
    font-family: 'Inter', sans-serif;
    margin: 0;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
}

.container {
    width: 100%;
    max-width: 540px;
    background-color: white;
    padding: 30px 35px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 15px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 24px;
    font-size: 24px;
}

.table-form {
    width: 100%;
    border-collapse: collapse;
    border: 1px solid #a3b8cc;
    margin-bottom: 20px;
    background-color: #f7faff;
}

.table-form th {
    background-color: #9db2e8;
    color: #1a237e;
    font-size: 15px;
    font-weight: 700;
    padding: 10px 14px;
    text-align: left;
    border: 1px solid #829bc9;
}

.table-form td {
    padding: 10px 14px;
    border: 1px solid #c5d3e8;
    vertical-align: top;
    font-size: 14px;
}

.label-cell {
    width: 38%;
    font-weight: 600;
    color: #222;
    background-color: #eef3fc;
}

.value-cell {
    width: 62%;
    background-color: #ffffff;
}

input[type="text"],
select,
textarea {
    width: 100%;
    padding: 8px 10px;
    border: 1px solid lightgray;
    border-radius: 4px;
```

```
    font-size: 14px;
    font-family: inherit;
    outline: none;
    transition: border-color 0.2s;
}

input[type="text"]:focus,
select:focus,
textarea:focus {
    border-color: royalblue;
}

textarea {
    resize: vertical;
}

.radio-group,
.checkbox-group {
    display: flex;
    flex-direction: column;
    gap: 8px;
}

.radio-item,
.checkbox-item {
    display: flex;
    align-items: center;
    gap: 8px;
    cursor: pointer;
    font-size: 14px;
}

.radio-item input,
.checkbox-item input {
    cursor: pointer;
}

.full-row-header {
    background-color: #eef3fc;
    font-weight: 600;
    color: #222;
    border: 1px solid #c5d3e8;
    padding: 10px 14px;
}

.textarea-cell {
    background-color: #ffffff;
    border: 1px solid #c5d3e8;
    padding: 10px 14px;
}

.action-row {
    background-color: #eef3fc;
    text-align: center;
    padding: 12px;
    border: 1px solid #c5d3e8;
}

#submitBtn {
    padding: 9px 24px;
    background-color: #e0e0e0;
    color: #111;
    border: 1px solid #999;
    border-radius: 4px;
    font-size: 14px;
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s;
}
```

```
#submitBtn:hover {
    background-color: #cfcfcf;
}

.error-msg {
    color: #d32f2f;
    font-size: 12px;
    margin-top: 4px;
    min-height: 14px;
}

.summary-card {
    display: none;
    margin-top: 20px;
    padding: 16px 20px;
    background-color: #f1f8e9;
    border: 1px solid #81c784;
    border-radius:  8px;
    color: #1b5e20;
    font-size: 14px;
    line-height: 1.6;
}

.summary-card h3 {
    margin-top: 0;
    margin-bottom: 10px;
    color: #2e7d32;
    font-size: 16px;
}

footer {
    text-align: center;
    margin-top: 30px;
    font-size: 14px;
    color: gray;
    border-top: 1px solid lightgray;
    padding-top: 15px;
}
```

Case Study 8.js

```
document.getElementById("name").addEventListener("input", function() {
    let val = this.value.trim();
    if (val === "") {
        document.getElementById("nameError").innerHTML = "Name is required.";
    } else if (!/^[A-Za-z ]+$/.test(val)) {
        document.getElementById("nameError").innerHTML = "Only letters allowed.";
    } else {
        document.getElementById("nameError").innerHTML = "";
    }
});

let sexRadios = document.getElementsByName("sex");
for (let i = 0; i < sexRadios.length; i++) {
    sexRadios[i].addEventListener("change", function() {
        document.getElementById("sexError").innerHTML = "";
    });
}

document.getElementById("ability").addEventListener("input", function() {
    let val = this.value.trim();
    if (val.length > 0) {
        document.getElementById("abilityError").innerHTML = "";
    }
});

document.getElementById("gymForm").addEventListener("submit",  function(e)  {
```

```
    e.preventDefault();

    let name = document.getElementById("name").value.trim();
    let sex = "";
    for (let i = 0; i < sexRadios.length; i++) {
        if (sexRadios[i].checked) {
            sex = sexRadios[i].value;
            break;
        }
    }
    let eyeColor = document.getElementById("eyeColor").value;
    let over6Feet = document.getElementById("over6Feet").checked;
    let over200Pounds = document.getElementById("over200Pounds").checked;
    let ability = document.getElementById("ability").value.trim();

    let isValid = true;

    if (name === "") {
        document.getElementById("nameError").innerHTML = "Name is required.";
        isValid = false;
    } else if (!/^[A-Za-z ]+$/.test(name)) {
        document.getElementById("nameError").innerHTML = "Only letters allowed.";
        isValid = false;
    } else {
        document.getElementById("nameError").innerHTML = "";
    }

    if (sex === "") {
        document.getElementById("sexError").innerHTML = "Please select your sex.";
        isValid = false;
    } else {
        document.getElementById("sexError").innerHTML = "";
    }

    if (ability === "") {
        document.getElementById("abilityError").innerHTML = "Please describe your athletic ability.";
        isValid = false;
    } else {
        document.getElementById("abilityError").innerHTML = "";
    }

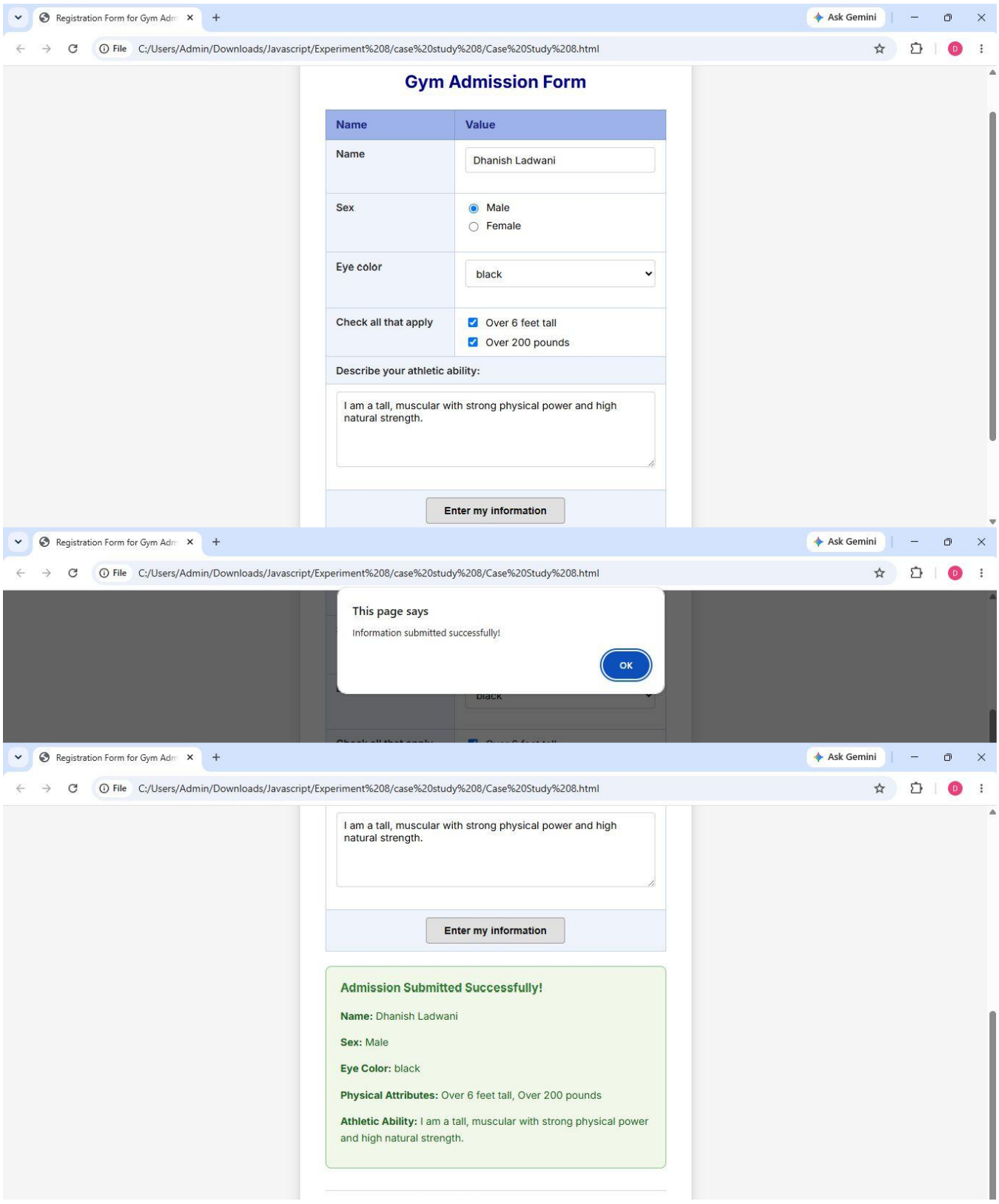
    let resultCard = document.getElementById("resultCard");

    if (isValid) {
        let appliedTraits = [];
        if (over6Feet) appliedTraits.push("Over 6 feet tall");
        if (over200Pounds) appliedTraits.push("Over 200 pounds");
        let traitsDisplay = appliedTraits.length > 0 ? appliedTraits.join(", ") : "None";

        resultCard.style.display = "block";
        resultCard.innerHTML = "<h3>Admission Submitted Successfully!</h3>" +
            "<p><b>Name:</b> " + name + "</p>" +
            "<p><b>Sex:</b> " + sex + "</p>" +
            "<p><b>Eye Color:</b> " + eyeColor + "</p>" +
            "<p><b>Physical Attributes:</b> " + traitsDisplay + "</p>" +
            "<p><b>Athletic Ability:</b> " + ability + "</p>";

        alert("Information submitted successfully!");
    } else {
        resultCard.style.display = "none";
        resultCard.innerHTML = "";
        alert("Please correct the errors before submitting.");
    }
});
```

4. Output:


Registration Form for Gym Adm
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%208/case%20study%208/Case%20Study%208.html
Gym Admission Form
Name
Value
Name
Dhanish Ladwani
Sex
Male
Female
Eye color
black
Check all that apply
Over 6 feet tall
Over 200 pounds
Describe your athletic ability:
I am a tall, muscular with strong physical power and high natural strength.
Enter my information
This page says
Information submitted successfully!
OK
Admission Submitted Successfully!
Name: Dhanish Ladwani
Sex: Male
Eye Color: black
Physical Attributes: Over 6 feet tall, Over 200 pounds
Athletic Ability: I am a tall, muscular with strong physical power and high natural strength.

5. Result/Conclusion:

A gym admission registration form was designed using appropriate regex-based validation and JavaScript event handling, ensuring correctly formatted entries for personal and membership details before submission. This case study reinforced applying form validation and events in a real-world registration scenario.